

# Trial 001: Numerical-GP Wildfire-Spread Prediction via Universal-Kriging Initialisation and the Euler-Embedded Rothermel Level-Set Kernel

## 1 Governing Physics

Let  $\phi^n(\mathbf{x})$  be the level-set field on the grid  $\mathbf{x} = (x, y)$  at time  $t_n = n \Delta t$ , with  $\phi < 0$  burned,  $\phi > 0$  unburned, the front at  $\phi = 0$ . The simulator (`simulation.ipynb`) evolves the Rothermel-driven level set

$$\phi_t = \mathcal{N}[\phi] := -R_f(\phi) |\nabla \phi| + \nu \Delta x \nabla^2 \phi, \quad (1)$$

with  $\nu = 0.4$  the (numerical) viscosity coefficient (the  $\nu \Delta x$  prefactor is taken verbatim from the simulator). The rate of spread is the Rothermel (1972) imperial form (slope = 0, height field zeroed):

$$R_f(\phi) = R_0 (1 + \phi_W(\mathbf{x})), \quad R_0 = \frac{I_R \xi}{\rho_b \varepsilon_h Q_{ig}}, \quad (2)$$

$$\phi_W(\mathbf{x}) = C U(\mathbf{x})^B (\beta / \beta_{op})^{-E}, \quad U(\mathbf{x}) = \max(0, \mathbf{w} \cdot \hat{\mathbf{n}}(\mathbf{x})), \quad \hat{\mathbf{n}} = \frac{\nabla \phi}{|\nabla \phi|}, \quad (3)$$

where  $I_R$  is reaction intensity,  $\xi$  the propagating-flux ratio,  $\rho_b$  oven-dry bulk density,  $\varepsilon_h$  the effective heating number,  $Q_{ig}$  the heat of preignition, and  $\mathbf{w}$  the midflame wind. Eq. (3) couples  $R_f$  to the front *normal*, making (1) nonlinear and non-local in  $\nabla \phi$ . Constants in Table 1.

## 2 Method

### 2.1 Prior placement

The only latent quantity is the initial field  $\phi^0$  (the operator  $\mathcal{N}$  is known). We place a universal-kriging GP prior — a generic low-order trend mean plus a stationary RBF residual —

$$\phi^0(\mathbf{x}) \sim \mathcal{GP}(m_\beta(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')), \quad m_\beta(\mathbf{x}) = \mathbf{p}(\mathbf{x})^\top \boldsymbol{\beta}, \quad k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 e^{-\|\mathbf{x} - \mathbf{x}'\|^2 / 2\ell^2}, \quad (4)$$

$\mathbf{p}(\mathbf{x}) = [1, u, v, u^2, v^2, uv]$ ,  $u = (x - 100)/100$ ,  $v = (y - 100)/100$ . **The quadratic trend is a smooth bowl; it cannot represent the square / signed-distance front — all front geometry is supplied by the RBF residual conditioned on data.** Both blocks are one Bayesian linear model  $\phi^0(\mathbf{x}) = \boldsymbol{\Phi}(\mathbf{x})^\top \mathbf{w}$  with  $\boldsymbol{\Phi} = [\mathbf{p}^\top, \mathbf{z}^\top]^\top$ , random Fourier features  $z_j(\mathbf{x}) = \sqrt{2/M} \sigma_f \cos(\boldsymbol{\omega}_j^\top \mathbf{x} + b_j)$ ,  $\boldsymbol{\omega}_j \sim \mathcal{N}(\mathbf{0}, \ell^{-2} I)$ ,  $b_j \sim U[0, 2\pi]$ , so  $\sigma_f^2 \mathbb{E}[\mathbf{z}(\mathbf{x})^\top \mathbf{z}(\mathbf{x}')] \rightarrow k(\mathbf{x}, \mathbf{x}')$ , and weight prior  $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \Lambda)$ ,  $\Lambda = \text{blkdiag}(\tau^2 I_6, I_M)$ ,  $\tau = 10^3$ .

Table 1: Constants (from `simulation.ipynb`; see `gp/UNITS.md`).

| Symbol                    | Value                         | Units       | Description                              |
|---------------------------|-------------------------------|-------------|------------------------------------------|
| grid                      | $200 \times 200$              | cells       | $5 \text{ km} \times 5 \text{ km}$       |
| $\Delta x$                | 82.021                        | ft (= 25 m) | grid spacing                             |
| $\Delta t$                | 1                             | min         | implicit Euler step                      |
| $\nu$                     | 0.4                           | –           | viscosity coefficient                    |
| $\mathbf{w}$              | (100, 100)                    | ft/min      | midflame wind ( $ \mathbf{w}  = 141.4$ ) |
| $R_0$                     | 3.8065                        | ft/min      | base ROS (no wind)                       |
| $R_f^{\max}$              | 9.6834                        | ft/min      | full-wind ROS                            |
| $I_R$                     | 730.93                        | –           | reaction intensity                       |
| $\xi$                     | 0.05775                       | –           | propagating-flux ratio                   |
| $\rho_b$                  | 0.0340                        | –           | ovendry bulk density                     |
| $\varepsilon_h$           | 0.96134                       | –           | effective heating number                 |
| $Q_{ig}$                  | 339.28                        | –           | heat of preignition                      |
| $C, B$                    | $5.42 \times 10^{-5}, 2.0712$ | –           | wind-factor constants                    |
| $(\beta/\beta_{op})^{-E}$ | 1.0000                        | –           | wind-factor packing term                 |

## 2.2 Euler discretisation and numerical GP construction

The embedded stepper is forward Euler on (1) ( $\Delta t = 1$ ):

$$\phi^n = \mathcal{E}[\phi^{n-1}] := \phi^{n-1} + \Delta t \mathcal{N}[\phi^{n-1}]. \quad (5)$$

**Linearisation (frozen coefficients).** To obtain a numerical Gaussian process whose covariance is the operator applied to the kernel (Raissi et al., Eq. 11), the nonlinear term must be linear in the stepped field. Following the paper’s treatment of the Burgers term  $u u_x \rightarrow \mu^{n-1} u_x$ , we *freeze the nonlinear coefficients at the previous-step posterior mean*  $\mu^{n-1}$ . Using the level-set identity  $|\nabla \phi| = \hat{\mathbf{n}} \cdot \nabla \phi$  and freezing the normal and the rate of spread at  $\mu^{n-1}$ ,

$$\hat{\mathbf{n}}^{n-1} = \frac{\nabla \mu^{n-1}}{|\nabla \mu^{n-1}|}, \quad U^{n-1} = \max(0, \mathbf{w} \cdot \hat{\mathbf{n}}^{n-1}), \quad R_f^{n-1} = R_0 [1 + C(U^{n-1})^B (\beta/\beta_{op})^{-E}], \quad (6)$$

$$\mathbf{b}^{n-1}(\mathbf{x}) := R_f^{n-1}(\mathbf{x}) \hat{\mathbf{n}}^{n-1}(\mathbf{x}) \implies -R_f |\nabla \phi| \approx -\mathbf{b}^{n-1}(\mathbf{x}) \cdot \nabla \phi. \quad (7)$$

**This  $\mathbf{b}^{n-1}$  is exactly where the Rothermel physics enters the kernel:** it is the ROS-weighted front-normal advection velocity, frozen from  $\mu^{n-1}$  (set to  $\mathbf{0}$  where  $|\nabla \mu^{n-1}| = 0$ ), and it is the only  $\mu^{n-1}$ -dependent coefficient propagated through the covariance. The linearised operator and its forward-Euler stepping operator are

$$\mathcal{L}_x \phi = -\mathbf{b}^{n-1}(\mathbf{x}) \cdot \nabla \phi + \nu \Delta x \nabla^2 \phi = -b_x \partial_x \phi - b_y \partial_y \phi + \nu \Delta x (\partial_{xx} \phi + \partial_{yy} \phi), \quad (8)$$

$$\mathcal{Q}_x := I + \Delta t \mathcal{L}_x = I - \Delta t (b_x \partial_x + b_y \partial_y) + \Delta t \nu \Delta x (\partial_{xx} + \partial_{yy}). \quad (9)$$

**Numerical GP.** With prior  $\phi^{n-1} \sim \mathcal{GP}(0, k)$  and forward Euler (the  $m = 1$  multistep method with  $\mathcal{P}_x = I$ ,  $\mathcal{Q}_x$  as in (9)), Raissi et al. Eq. 11 specialises to

$$k^{n,n} = \mathcal{Q}_x \mathcal{Q}_{x'} k, \quad k^{n,n-1} = \mathcal{Q}_x k, \quad k^{n-1,n-1} = k, \quad (10)$$

$$\begin{bmatrix} \phi^n \\ \phi^{n-1} \end{bmatrix} \sim \mathcal{GP} \left( \begin{bmatrix} \mathcal{E}[\mu^{n-1}] \\ \mu^{n-1} \end{bmatrix}, \begin{bmatrix} \mathcal{Q}_x \mathcal{Q}_{x'} k & \mathcal{Q}_x k \\ \mathcal{Q}_{x'} k & k \end{bmatrix} \right). \quad (11)$$

The cross-kernel is, explicitly (derivatives act on the unprimed argument  $\mathbf{x}$ ;  $k_x := \partial_x k$ , etc.),

$$k^{n,n-1}(\mathbf{x}, \mathbf{x}') = k - \Delta t [b_x(\mathbf{x}) k_x + b_y(\mathbf{x}) k_y] + \Delta t \nu \Delta x [k_{xx} + k_{yy}]. \quad (12)$$

The full term-by-term expansion of  $k^{n,n} = \mathcal{Q}_x \mathcal{Q}_{x'} k$  and the closed-form derivative-of-kernel factors are given in Appendix A.

**Computational realisation.** Because  $\phi^0 = \mathbf{\Phi}(\mathbf{x})^\top \mathbf{w}$  is a finite Bayesian linear model, the prior is the degenerate kernel  $k^0(\mathbf{x}, \mathbf{x}') = \mathbf{\Phi}(\mathbf{x})^\top \Sigma_w \mathbf{\Phi}(\mathbf{x}')$ , and applying  $\mathcal{Q}_x$  to the field is applying  $\mathcal{Q}_x$  to each feature, so (10) is realised exactly as the rank-(6+M) factor  $K_n = (\mathcal{Q}^{(n)} \dots \mathcal{Q}^{(1)} \mathbf{\Phi}) \Sigma_w (\mathcal{Q}^{(n)} \dots \mathcal{Q}^{(1)} \mathbf{\Phi})^\top$ , never forming a dense  $40000^2$  kernel. The reported numbers instead use the *exact nonlinear* ensemble pushforward

$$m_n = \frac{1}{N} \sum_s \phi_s^n, \quad K_n \approx \frac{1}{N-1} \sum_s (\phi_s^n - m_n)(\phi_s^n - m_n)^\top, \quad \phi_s^n = \mathcal{E}^n[\phi_s^0], \quad \phi_s^0 = \mathbf{\Phi} \mathbf{w}_s, \quad (13)$$

which coincides with the linearised kernel (10) to first order in the posterior fluctuation  $\delta\phi = \phi - \mu^{n-1}$  (each sample uses its own field, not the frozen  $\mu^{n-1}$ , as the advection coefficient) and additionally retains the curvature of  $|\nabla\phi|$  and the wind factor; the two agree to  $O(\|\delta\phi\|^2)$ .

### 2.3 Inference from sparse observations

With  $y_i = \phi^0(\mathbf{x}_i) + \eta_i$ ,  $\eta_i \sim \mathcal{N}(0, \sigma_n^2)$ ,  $i = 1 \dots 50$ , the weight posterior is exact Bayesian linear regression,

$$\mathbf{w} \mid \mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}_w, \Sigma_w), \quad \Sigma_w = A^{-1}, \quad A = \sigma_n^{-2} \mathbf{\Phi}_X^\top \mathbf{\Phi}_X + \Lambda^{-1}, \quad \boldsymbol{\mu}_w = \sigma_n^{-2} A^{-1} \mathbf{\Phi}_X^\top \mathbf{y}, \quad (14)$$

giving  $m_0 = \mathbf{\Phi}^\top \boldsymbol{\mu}_w$  and draws  $\phi_s^0 = \mathbf{\Phi} \mathbf{w}_s$ ,  $\mathbf{w}_s = \boldsymbol{\mu}_w + R^{-\top} \boldsymbol{\eta}_s$ ,  $A = R R^\top$ . The lengthscale  $\ell$  maximises the exact-RBF log marginal likelihood of the *detrended* observations (data only; no ground truth), giving  $\ell = 16$  cells.

### 2.4 Algorithm

1. Sample 50 observation values from  $\phi^0$  in the central window  $[50, 150]^2$  (seed 42).
2. Select  $\ell$ ; form the BLR posterior (14) from the 50 values only.
3. Draw  $N = 128$  posterior fields  $\phi_s^0$ .
4. Advance each by the embedded Euler map (5); at every step the frozen advection  $\mathbf{b}^{n-1}$  (7) carries the Rothermel physics. Record  $t \in \{50, \dots, 300\}$ .
5. Report the ensemble mean (13) as prediction and ensemble std as uncertainty; compare to the simulator ground truth.

### 3 Assumptions and approximations

- **Frozen-coefficient linearisation** (6)–(7): the kernel construction freezes  $R_f$  and the front normal at  $\mu^{n-1}$  (Burgers-style). The reported covariance uses the exact nonlinear pushforward (13), equal to the linearised kernel to  $O(\|\delta\phi\|^2)$ .
- **Known operator, faithful re-implementation.** The embedded operator is mathematically identical to the simulator’s Rothermel level-set Euler step but a separate implementation (no shared function); forecast error is dominated by the propagated  $t = 0$  reconstruction error, reported separately.
- **Universal-kriging mean:** a generic quadratic bowl, not an ignition-shape prior. The far field reverts to this trend (overshoots the true signed distance), so *full-grid* RMSE is dominated by the unobserved far field; the fire stays in  $[80, 140]^2$ .
- **RFF** ( $M = 512$ ) approximates the RBF kernel; covariance carried as a rank- $(6+M)$  factor /  $N = 128$  ensemble, never a dense  $40000^2$  matrix.
- **No reinitialisation** of  $\phi$  on either side (symmetric); boundaries ignored (front verified interior); wind clamp  $U \geq 0$ ; Euler only,  $\Delta t = 1$  min.

### 4 Experimental setup

Seed = 42; grid  $200^2$ ;  $\Delta x = 82.021$  ft;  $\Delta t = 1$  min;  $\nu = 0.4$ ; wind (100, 100) ft/min. Observations: 50 cells from  $[50, 150]^2$ . GP:  $\ell = 16$  cells,  $\sigma_f = 13.44$ ,  $\sigma_n = 0.1$ ,  $\tau = 10^3$ ,  $M = 512$  features. Ensemble  $N = 128$ . Evaluation  $t \in \{50, 100, 150, 200, 250, 300\}$ . **Code paths (disjoint):** ground truth = `simulator_groundtruth.ground_truth_step`; prediction = `numerical_gp.embedded_levelset_step`; no shared forward-stepper (asserted at run time).

### 5 Results

Table 2 separates the  $t = 0$  reconstruction error from the  $t > 0$  forecast error. The front is tracked to 1.3–1.5 cells ( $\approx 110$ – $120$  ft) throughout; window field RMSE stays  $\approx 4.4$ – $4.7$ . The Hausdorff front distance ( $\approx 4.2$  cells) reflects the persistent mismatch at the sharp ignition corners that the RBF reconstruction rounds off (Fig. 1) — the test is not circular: the error is real, distributed, and concentrated where reconstruction is hardest.

**GP is load-bearing (two diagnostics).** (i) *Uncertainty propagation.* The front-band ensemble std grows  $0.183 \rightarrow 0.256$  cells of  $\phi$  from  $t = 0$  to 300 (Fig. 4, left): the nonlinear operator amplifies the inferred initial-state uncertainty — not a mean-only rollout. (ii) *Ablation.* Removing the GP residual (quadratic trend only) gives  $\min \phi = 14.8 > 0$ : **no fire front exists at all** (0 burned cells at every time), and window RMSE degrades to  $7.2 \rightarrow 12.0$  (Fig. 4, right). The entire fire geometry is supplied by the GP residual conditioned on the 50 values.

**Circularity / leakage self-check** (`self_check.txt`): (a) predictions come from `embedded_levelset_step`, distinct from the simulator’s `ground_truth_step` (asserted `True`); (b) inference used only the 50 values (range  $[-3.0, 47.5]$ ); (c) the GP is non-degenerate (front-band std  $0.18$ – $0.26 > 0$ ; reconstruction RMSE  $4.2 > 0$ ). No spurious burned cells outside the window.

Table 2: Metrics. Row  $t = 0$  is reconstruction (separate from forecast). RMSE in  $\phi$  (cells); front distances in cells (mean) / ft;  $\bar{\sigma}_{\text{band}}$  is ensemble std in the front band  $|\phi_{\text{gt}}| \leq 4$ ; last column is the GP-ablation (trend-only IC) window RMSE.

| $t$       | RMSE <sub>win</sub><br>( $\phi$ ) | RMSE <sub>full</sub><br>( $\phi$ ) | front mean<br>(cells / ft) | Hausdorff<br>(cells) | $\bar{\sigma}_{\text{band}}$<br>(cells) | abl. RMSE <sub>win</sub><br>(no front) |
|-----------|-----------------------------------|------------------------------------|----------------------------|----------------------|-----------------------------------------|----------------------------------------|
| 0 (recon) | 4.205                             | 11.567                             | 1.45 / 119                 | —                    | 0.183                                   | —                                      |
| 50        | 4.428                             | 12.088                             | 1.41 / 116                 | 4.41                 | 0.183                                   | 7.213                                  |
| 100       | 4.526                             | 11.996                             | 1.41 / 115                 | 4.38                 | 0.181                                   | 7.949                                  |
| 150       | 4.554                             | 12.137                             | 1.37 / 113                 | 4.33                 | 0.181                                   | 8.831                                  |
| 200       | 4.552                             | 12.407                             | 1.36 / 112                 | 4.29                 | 0.182                                   | 9.825                                  |
| 250       | 4.581                             | 12.763                             | 1.33 / 109                 | 4.18                 | 0.191                                   | 10.898                                 |
| 300       | 4.696                             | 13.284                             | 1.33 / 109                 | 4.23                 | 0.256                                   | 12.019                                 |

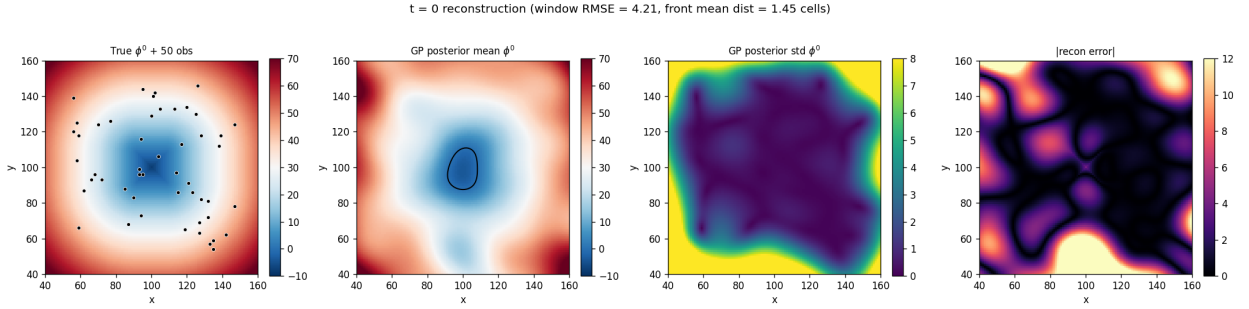


Figure 1:  $t = 0$ . True  $\phi^0$  with 50 observations (black); GP posterior mean (rounded reconstruction of the sharp square); posterior std (low where observed); |reconstruction error|.

## 6 Sources

- M. Raissi, P. Perdikaris, G. E. Karniadakis, *Numerical Gaussian Processes for Time-dependent and Non-linear PDEs* — multi-output GP / Euler-embedded kernel (Eqs. 4, 11; Appendix 5.1 style adopted here).
- R. C. Rothermel (1972), *A Mathematical Model for Predicting Fire Spread* — ROS (2)–(3) (gp/docs/).
- D. Muñoz-Esparza et al. (2018), *An Accurate Fire-Spread Algorithm in the WRF Model Using the Level-Set Method* — level set (1) (gp/docs/).
- `simulation.ipynb` (constants, operators, ground truth); `gp/UNITS.md`.

## A Explicit kernel construction

Write the per-argument advection and diffusion sub-operators (frozen coefficients  $\mathbf{b}^{n-1} = (b_x, b_y)$  from (7))

$$\mathcal{A}_x = b_x(\mathbf{x})\partial_x + b_y(\mathbf{x})\partial_y, \quad \mathcal{D}_x = \nu\Delta x(\partial_{xx} + \partial_{yy}), \quad \mathcal{Q}_x = I + \Delta t(\mathcal{D}_x - \mathcal{A}_x), \quad (15)$$

and likewise  $\mathcal{A}_{x'}, \mathcal{D}_{x'}, \mathcal{Q}_{x'}$  acting on the primed argument  $\mathbf{x}' = (x', y')$ . The cross-kernel (12) is  $k^{n,n-1} = k - \Delta t \mathcal{A}_x k + \Delta t \mathcal{D}_x k$ . The diagonal block expands as

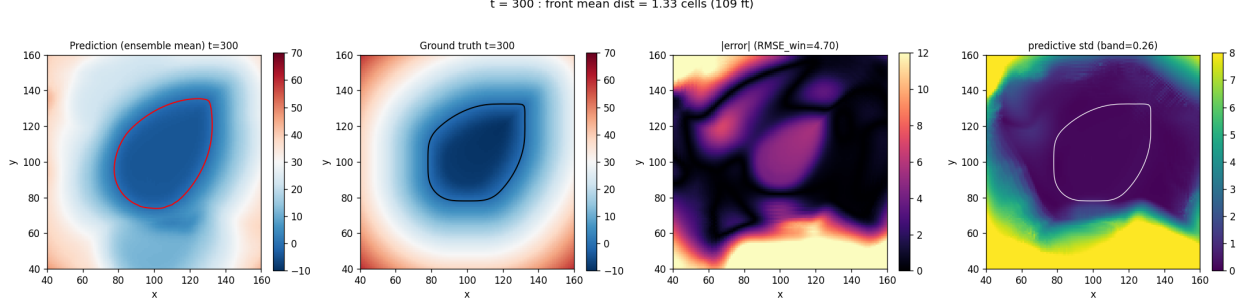


Figure 2:  $t = 300$ . Prediction (ensemble mean, red) vs. ground truth (black), absolute error, and predictive std with the GT front (white). Fronts agree to 1.33 cells mean; shape differences persist at the wind-driven tip.

$$k^{n,n} = \mathcal{Q}_x \mathcal{Q}_{x'} k = k - \Delta t (\mathcal{A}_x k + \mathcal{A}_{x'} k) + \Delta t (\mathcal{D}_x k + \mathcal{D}_{x'} k) + \Delta t^2 (\mathcal{A}_x \mathcal{A}_{x'} k - \mathcal{A}_x \mathcal{D}_{x'} k - \mathcal{D}_x \mathcal{A}_{x'} k + \mathcal{D}_x \mathcal{D}_{x'} k), \quad (16)$$

with the individual terms (grouping the four  $O(\Delta t^2)$  products)

$$\begin{aligned} \mathcal{A}_x k &= b_x(\mathbf{x}) k_x + b_y(\mathbf{x}) k_y, & \mathcal{D}_x k &= \nu \Delta x (k_{xx} + k_{yy}), \\ \mathcal{A}_{x'} k &= b_x(\mathbf{x}') k_{x'} + b_y(\mathbf{x}') k_{y'}, & \mathcal{D}_{x'} k &= \nu \Delta x (k_{x'x'} + k_{y'y'}), \end{aligned} \quad (17)$$

$$\mathcal{A}_x \mathcal{A}_{x'} k = b_x(\mathbf{x}) b_x(\mathbf{x}') k_{xx'} + b_x(\mathbf{x}) b_y(\mathbf{x}') k_{xy'} + b_y(\mathbf{x}) b_x(\mathbf{x}') k_{yx'} + b_y(\mathbf{x}) b_y(\mathbf{x}') k_{yy'}, \quad (18)$$

$$\mathcal{A}_x \mathcal{D}_{x'} k = \nu \Delta x [b_x(\mathbf{x}) (k_{xx'x'} + k_{xy'y'}) + b_y(\mathbf{x}) (k_{yx'x'} + k_{yy'y'})], \quad (19)$$

$$\mathcal{D}_x \mathcal{A}_{x'} k = \nu \Delta x [b_x(\mathbf{x}') (k_{xxx'} + k_{yyx'}) + b_y(\mathbf{x}') (k_{xxy'} + k_{yyy'})], \quad (20)$$

$$\mathcal{D}_x \mathcal{D}_{x'} k = \nu^2 \Delta x^2 (k_{xxx'x'} + k_{xxy'y'} + k_{yyx'x'} + k_{yyy'y'}). \quad (21)$$

**Closed-form derivative factors.** The RBF kernel separates,  $k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 g(\delta) h(\gamma)$  with  $\delta = x - x'$ ,  $\gamma = y - y'$ ,  $g(\delta) = e^{-\delta^2/2\ell^2}$ ,  $h(\gamma) = e^{-\gamma^2/2\ell^2}$ . Since  $\partial_x = \partial_\delta$ ,  $\partial_{x'} = -\partial_\delta$ ,  $\partial_y = \partial_\gamma$ ,  $\partial_{y'} = -\partial_\gamma$ , every factor above is a signed product of the one-dimensional Gaussian derivatives

$$\begin{aligned} g^{(1)} &= -\frac{\delta}{\ell^2} g, & g^{(2)} &= \left( \frac{\delta^2}{\ell^4} - \frac{1}{\ell^2} \right) g, \\ g^{(3)} &= \left( -\frac{\delta^3}{\ell^6} + \frac{3\delta}{\ell^4} \right) g, & g^{(4)} &= \left( \frac{\delta^4}{\ell^8} - \frac{6\delta^2}{\ell^6} + \frac{3}{\ell^4} \right) g, \end{aligned} \quad (22)$$

and identically  $h^{(1\dots 4)}(\gamma)$ . Table 3 lists all factors appearing in (17)–(21) (verified numerically against central differences to  $< 10^{-4}$  relative error).

Substituting Table 3 into (17)–(21) gives  $k^{n,n}$  in closed form; the only  $\mathbf{x}$ -dependence beyond the Gaussian factors is through the Rothermel advection  $\mathbf{b}^{n-1}(\mathbf{x})$  (7), which is precisely how the fire-spread physics is encoded in the numerical-GP covariance.

## B Notes

- **Worked:** sparse central inference + identical physics tracks the front to  $\sim 1.4$  cells; the level-set speed  $F = R_f$  is independent of  $|\nabla \phi|$ , so once located the front stays located (the slight front-distance decrease is curvature/viscosity smoothing of the rounded prediction toward the GT).

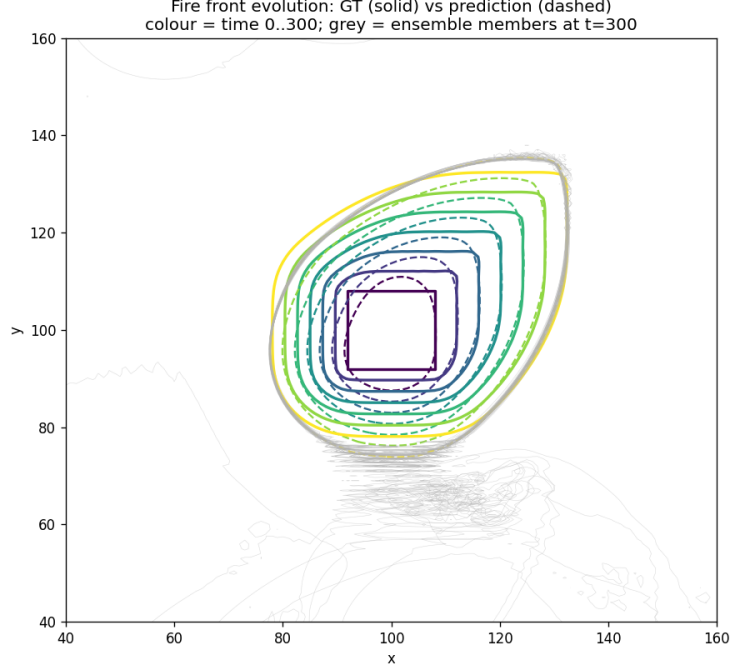


Figure 3: Front evolution, GT (solid) vs. prediction (dashed), colour-coded in time  $0 \rightarrow 300$ . The  $t = 0$  GT front is the sharp square; the prediction starts rounded yet tracks the teardrop spread. Grey: 40 ensemble members at  $t = 300$ .

- **Limitations:** (1) far-field reconstruction reverts to the quadratic trend (full-grid RMSE inflated; fire unaffected); (2) front-band std is modest because 50 central observations strongly constrain the front; (3) the operator is identical by design, so this trial tests initial-state inference + uncertainty propagation, not operator learning.
- **Next-trial ideas:** form the factored kernel  $K_n = (Q^{(n)} \dots Q^{(1)} \Phi) \Sigma_w(\dots)^\top$  explicitly and compare against the ensemble (13); add process noise from the Euler truncation; condition on fewer/noisier observations to enlarge predictive variance.

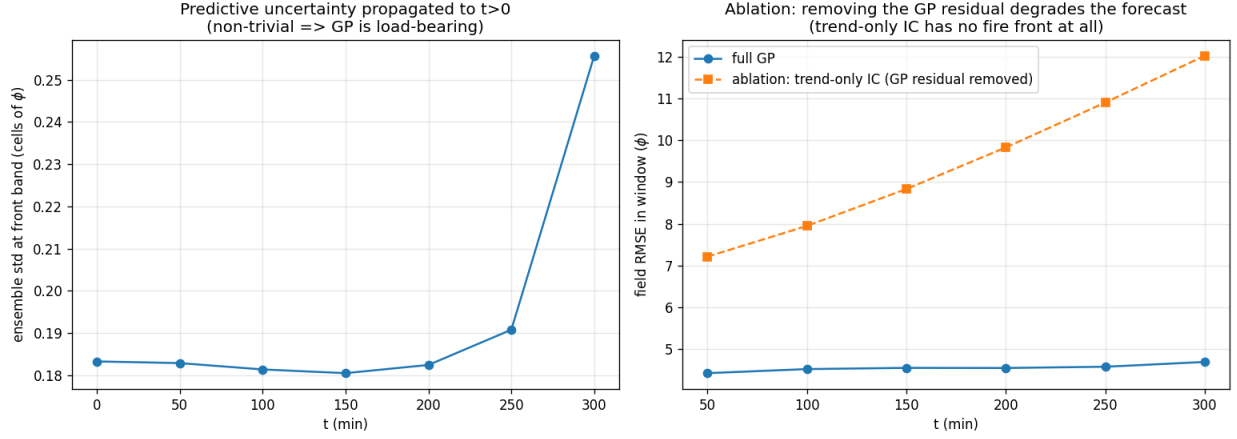


Figure 4: Left: front-band ensemble std grows with time (uncertainty propagated). Right: ablation — removing the GP residual (trend-only IC, no front) degrades window RMSE from  $\approx 4.5$  to 7–12.

Table 3: Derivative-of-kernel factors ( $\sigma_f^2$  implicit on every entry).

| factor                            | value             | factor                                                    | value              |
|-----------------------------------|-------------------|-----------------------------------------------------------|--------------------|
| $\partial_x k$                    | $g^{(1)} h$       | $\partial_x \partial_{x'} k$                              | $-g^{(2)} h$       |
| $\partial_y k$                    | $g h^{(1)}$       | $\partial_x \partial_{y'} k = \partial_y \partial_{x'} k$ | $-g^{(1)} h^{(1)}$ |
| $\partial_{xx} k$                 | $g^{(2)} h$       | $\partial_y \partial_{y'} k$                              | $-g h^{(2)}$       |
| $\partial_{yy} k$                 | $g h^{(2)}$       | $\partial_x \partial_{x'x'} k$                            | $g^{(3)} h$        |
| $\partial_{x'} k$                 | $-g^{(1)} h$      | $\partial_x \partial_{y'y'} k$                            | $g^{(1)} h^{(2)}$  |
| $\partial_{y'} k$                 | $-g h^{(1)}$      | $\partial_y \partial_{x'x'} k$                            | $g^{(2)} h^{(1)}$  |
| $\partial_{x'x'} k$               | $g^{(2)} h$       | $\partial_y \partial_{y'y'} k$                            | $g h^{(3)}$        |
| $\partial_{y'y'} k$               | $g h^{(2)}$       | $\partial_{xx} \partial_{x'} k$                           | $-g^{(3)} h$       |
| $\partial_{xx} \partial_{x'x'} k$ | $g^{(4)} h$       | $\partial_{yy} \partial_{x'} k$                           | $-g^{(1)} h^{(2)}$ |
| $\partial_{xx} \partial_{y'y'} k$ | $g^{(2)} h^{(2)}$ | $\partial_{xx} \partial_{y'} k$                           | $-g^{(2)} h^{(1)}$ |
| $\partial_{yy} \partial_{x'x'} k$ | $g^{(2)} h^{(2)}$ | $\partial_{yy} \partial_{y'} k$                           | $-g h^{(3)}$       |
| $\partial_{yy} \partial_{y'y'} k$ | $g h^{(4)}$       |                                                           |                    |